C. Method of Inversion :-

1. If $y = \cos^{-1} x$, P.T. $x = 1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \cdots$

→ Tut

We have, $y = \cos^{-1} x$

$\therefore x = \cos y$

$\therefore x = 1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \cdots$

2. If $y = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$ P.T. $x = y - \frac{y^2}{2} + \frac{y^3}{3} - \frac{y^4}{4}$

→

Here, $y = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$

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$\therefore y = e^x - 1$

$\therefore y + 1 = e^x$

$\therefore x = \log(1+y)$

$\therefore x = y - \frac{y^2}{2} + \frac{y^3}{3} - \frac{y^4}{4} + \cdots$

d. Method of differentiation or integration of known series:

1. Expand $\tan^{-1}x$ upto the first four terms by Maclaurins series

→

Let, $y = \tan^{-1}x$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

$$= (1+x^2)^{-1}$$

$$= 1 - x^2 + x^4 - x^6 + \cdots \quad [\text{Binomial thm.}]$$

$\therefore dy = (1 - x^2 + x^4 - x^6 + \cdots)\,dx.$

Integrating both the sides,

$$y = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + c$$

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When $x=0$, $y=\tan^{-1}0=0$

$\therefore c=0$

$$\therefore y = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

$$\therefore \tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

2. Obtain the Maclaurin's series expansion for $\sqrt{1+\sin x}$.

→

Here, $\sqrt{1+\sin x} = \sqrt{1+2\cdot\sin\frac{x}{2}\cos\frac{x}{2}}$

$$= \sqrt{\sin^2\frac{x}{2}+\cos^2\frac{x}{2}+2\sin\frac{x}{2}\cos\frac{x}{2}}$$

$$= \sqrt{\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2}$$

$$= \left|\sin\frac{x}{2}+\cos\frac{x}{2}\right|$$

Replace x by $\frac{x}{2}$ in the expansion of $\sin x$

$$\therefore \sin\frac{x}{2} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\sin\frac{x}{2} = \frac{x}{2} - \frac{\left(\frac{x}{2}\right)^3}{3!} + \frac{\left(\frac{x}{2}\right)^5}{5!} - \frac{\left(\frac{x}{2}\right)^7}{7!} + \cdots$$

$$= \frac{x}{2} - \frac{x^3}{2^3\cdot 3!} + \frac{x^5}{2^5\cdot 5!} - \frac{x^7}{2^7\cdot 7!} + \cdots$$

Differentiating both the sides of this expansion,

$$\cos\frac{x}{2}\cdot\frac{1}{2} = \frac{1}{2} - \frac{2x^2}{2^3\cdot 3!} + \frac{5x^4}{2^5\cdot 5!} - \frac{7x^6}{2^7\cdot 7!} + \cdots$$

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$$\cos x/2 = 2\left(\frac{1}{2} - \frac{x^2}{2^3.2!} + \frac{x^4}{2^5.4!} - \frac{x^6}{2^7.6!} + \cdots\right)$$

$$= 1 - \frac{x^2}{2^2.2!} + \frac{x^4}{2^4.4!} - \frac{x^6}{2^6.6!} + \cdots$$

e. Method of substitution:-

Sometimes complicated functions can be reduced to the standard form by a suitable substitution.

1. P.T. $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right)$

(V.t)

We have $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

Let, $x = \tan\theta$

$$\therefore \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

$$= \sin^{-1}\left(\frac{2\tan\theta}{\sec^2\theta}\right)$$

$$= \sin^{-1}(2\sin\theta\cos\theta)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$= 2\theta$$

$$= 2\tan^{-1}x$$

$$= 2\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right)$$

[By Example (1) of d]

Then $\sqrt{1+\sin x} = \sin\frac{x}{2} + \cos\frac{x}{2}$

$= 1 + \frac{x}{2} - \frac{x^2}{2^2.2!} - \frac{x^3}{2^3.3!} + \frac{x^4}{2^4.4!} + \frac{x^5}{2^5.5!} - \cdots$

2. P.T. $\cos^{-1}[\tanh(\log x)] = \pi - 2\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right)$

→

Since, $\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$

$\tanh(\log x) = \frac{e^{\log x} - e^{-\log x}}{e^{\log x} + e^{-\log x}}$

$= \frac{x - x^{-1}}{x + x^{-1}} = \frac{x - \frac{1}{x}}{x + \frac{1}{x}}$

$= \frac{x^2 - 1}{x^2 + 1}$

$\therefore \cos^{-1}[\tanh(\log x)] = \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right)$

Now, Let $x = \tan\theta$

$\therefore \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) = \cos^{-1}\left(\frac{\tan^2\theta - 1}{\tan^2\theta + 1}\right)$

$= \cos^{-1}(-\cos 2\theta)$ Converting in $\sin\theta, \cos\theta$

$= \cos^{-1}[\cos(\pi - 2\theta)]$

$= \pi - 2\theta$

$= \pi - 2\tan^{-1}x$

$= \pi - 2\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right)$ [Ex. (1) of d]

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f Using complex numbers:-

1. Show that

i) $e^{ix} = \cos x + i \sin x$

ii) $e^{-ix} = \cos x - i \sin x$

iii) $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$

iv) $\cos x = \frac{e^{ix} + e^{-ix}}{2}$, where $i = \sqrt{-1}$.

→

Since, $e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \dots$,

where $z = x + iy$

Replacing z by (ix) in the above expansion,

$$e^{ix} = 1 + (ix) + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \dots$$

$$= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} - \frac{ix^5}{5!} + \dots$$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)$$

But $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$\therefore e^{ix} = \cos x + i \sin x$ — (1)

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Replacing x by $-x$ in (1),

$$e^{-ix} = \cos x - i \sin x \quad \text{——— (2)}$$

(1) − (2),

$$e^{ix} - e^{-ix} = -2i \sin x$$

$$\therefore \sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad \text{——— (3)}$$

(1) + (2)

$$e^{ix} + e^{-ix} = 2\cos x$$

$$\therefore \cos x = \frac{e^{ix} + e^{-ix}}{2} \quad \text{——— (4)}$$

2. P.T.

i. $e^{x} \sin x = x + x^2 + \frac{x^3}{3} - \frac{x^5}{30} - \frac{x^6}{90} - \cdots$

ii $e^{x} \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} - \frac{x^5}{30} - \cdots$

→

$$e^{x} \cdot e^{ix} = e^{x}(\cos x + i \sin x) = e^{(1+i)x}$$

Expanding $e^{(1+i)x}$ by expansion of e^{x},

$$e^{(1+i)x} = 1 + (1+i)x + \frac{(1+i)^2 x^2}{2!} + \frac{(1+i)^3 x^3}{3!} + \frac{(1+i)^4 x^4}{4!} + \cdots$$

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Now, $(1+i)^2 = 1+2i+i^2 = 2i$

$(1+i)^3 = 1^3+i^3+3i(1+i) = 1-i+3i+3i^2$

$= 1+2i-3$

$= 2i-2$

$(1+i)^4 = (1+i)^2(1+i)^2 = (2i)(2i) = -4$

$\therefore e^{(1+i)x} = 1+(1+i)x + 2i\frac{x^2}{2!} + (2i-2)\frac{x^3}{3!} - 4\frac{x^4}{4!} + \cdots$

$= 1+x+ix+\frac{2}{2!}ix^2 + 2(i-1)\frac{x^3}{3!} - 4\frac{x^4}{4!} + \cdots$

$= \left(1+x-\frac{2}{3!}x^3-\frac{4}{4!}x^4+\cdots\right) + i\left(x+\frac{2}{2!}x^2+\frac{2}{3!}x^3+\cdots\right)$

$\therefore e^x(\cos x + i\sin x) = \left(1+x-\frac{x^3}{3}-\frac{x^4}{6}+\cdots\right) + i\left(x+x^2+\frac{x^3}{3}+\cdots\right)$

- Equating real and imaginary parts,

$e^x \cos x = 1+x-\frac{x^3}{3}-\frac{x^4}{4}+\cdots$

$e^x \sin x = x+x^2+\frac{x^3}{3}+\cdots$

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g Using Leibnitz's theorem :-

1. Expand $\log(x+\sqrt{x^2+1})$ upto the first four terms by Maclaurin's theorem.

→

Let, $y = \log(x+\sqrt{x^2+1})$ ——— (1)

$$y_1 = \frac{1}{x+\sqrt{x^2+1}}\left[1+\frac{1}{2}(x^2+1)^{-\frac{1}{2}}(2x)\right]$$

$$= \frac{1}{x+\sqrt{x^2+1}}\left[\frac{1+\frac{x}{\sqrt{x^2+1}}}{}\right]$$

$$= \frac{1}{x+\sqrt{x^2+1}}\left[\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}}\right]$$

$$= \frac{1}{\sqrt{x^2+1}} \quad \text{——— (2)}$$

$\therefore \sqrt{x^2+1}\, y_1 = 1$

$\therefore (x^2+1) y_1^2 = 1$

Differentiating again,

$(x^2+1)\cdot 2y_1 y_2 + (2x) y_1^2 = 0$

Here, $2y_1 \neq 0$

$\therefore$ Dividing by $2y_1$, $(1+x^2) y_2 + x y_1 = 0$ ——— (3)

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Differentiating n times by Leibnitz's thm,

$[(x^2+1)y_{n+2} + {}^nC_1 y_{n+1}(2x) + {}^nC_2 y_n(2)] + [x y_{n+1} + {}^nC_1 y_n(1)] = 0$

→ Putting $x=0$ and simplifying,

$y_{n+2}(0) + 2{}^nC_2 y_n(0) + {}^nC_1 y_n(0) = 0$

$\therefore y_{n+2}(0) + 2\cdot\frac{n(n-1)}{2} y_n(0) + n\cdot y_n(0) = 0$

$\therefore y_{n+2}(0) + (n^2-n) y_n(0) + n y_n(0) = 0$

$\therefore y_{n+2}(0) = -n^2 y_n(0)$ ——— (4)

→ Putting $x=0$ in (1), (2) and (3),

$y=0,\ y_1=1,\ y_2=0$

→ Putting $n=1,2,3,4$ successively in (4),

when $x=0$

$y_3(0) = -y_1(0) = -1^2$

$y_4(0) = -2^2 y_2 = 0$

$y_5(0) = -3^2 \cdot y_3 = -3^2(-1^2) = 3^2 \cdot 1^2$

$y_6 = -4^2 \cdot y_4 = 0$

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$$\therefore y = (y)_0 + x(y_1)_0 + \frac{x^2}{2!}(y_2)_0 + \frac{x^3}{3!}(y_3)_0 + \cdots$$

$$= 0 + x(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-1^2) + \frac{x^4}{4!}(0) +$$

$$\frac{x^5}{5!}(32 \cdot 12) + \frac{x^6}{6!}(0) + \cdots$$

$$\therefore \log(x + \sqrt{1+x^2}) = x - \frac{x^3}{6} + \frac{3}{40}x^5 - \frac{5}{112}x^7 + \cdots$$

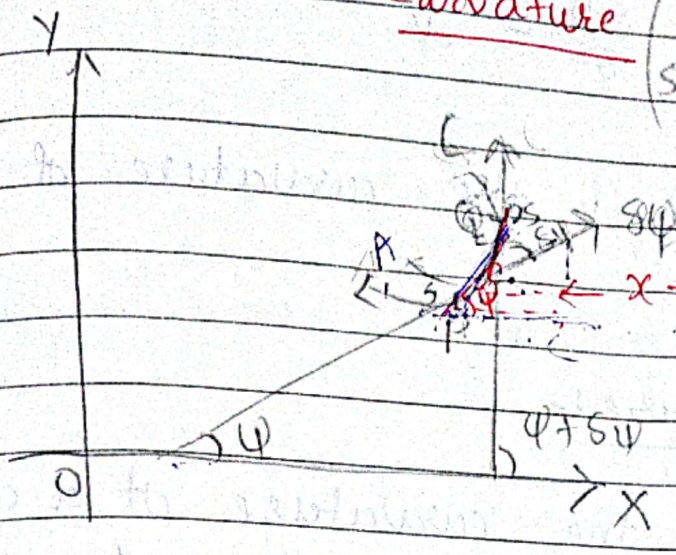
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R. Angle = length

$\delta\psi \cdot R = \delta s \Rightarrow R = \frac{\delta s}{\delta\psi} \Rightarrow \frac{1}{R} = \frac{\delta\psi}{\delta s}$

Curvature

(It measures degree of sharpness of bending of the curve)



Definition:-

Curvature:-

- Let, P be a point on a given curve and let Q be its neighbouring point.

- Let, Arc AP = s and Arc AQ = s + δs. So that Arc PQ = δs.

- Let, the tangents at P and Q i.e. makes an angles ψ and ψ + δψ resp. with the x-axis so that the angle between the two tangents is δψ.

- δψ denotes the bending or change in inclination as a point moves along the curve from P to Q through a distance δs. This suggests the following definitions:

i. This angle δψ is called the total bending or total curvature of the arc PQ.

ii. The rate $\frac{\delta\psi}{\delta s}$ is called the average curvature of the arc

iii. The curvature of the curve at P is defined as

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$$\lim_{Q \to P \text{ arc } PQ} \frac{\delta\psi}{\text{arc } PQ} = \lim_{\delta s \to 0} \frac{\delta\psi}{\delta s} = \frac{d\psi}{ds}.$$

∴ By definition, $\frac{d\psi}{ds}$ is the curvature of the curve at pt. P.

* Radius of curvature :-

The reciprocal of the curvature of a curve at any point is called its radius of curvature at the point and is generally denoted by ρ.

$$\therefore \rho = \frac{ds}{d\psi}.$$

Thus, the radius of curvature of a circle at any point is equal to the radius of the circle.

* Examples :-

1. Find the radius of curvature at the point (s, ψ) on the following curves.
   1. $s = c \tan\psi$ (catenary)
   2. $s = a \cot\psi \operatorname{cosec}\psi + a \log(\cot\psi + \operatorname{cosec}\psi)$ (parabola)

→

1. $s = c \tan\psi$.

Diff. w.r.t. ψ,

$$\frac{ds}{d\psi} = c \sec^2\psi$$

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$\therefore \rho = \frac{ds}{d\psi} = c \sec^2\psi$.

2. $s = a \cot\psi \operatorname{cosec}\psi + a \log(\cot\psi + \operatorname{cosec}\psi)$

$$\frac{ds}{d\psi} = +a\cot\psi(-\operatorname{cosec}\psi\cdot\cot\psi) + a(-\operatorname{cosec}^2\psi)\cdot\operatorname{cosec}\psi + a\cdot\frac{1}{\cot\psi+\operatorname{cosec}\psi}[-\operatorname{cosec}^2\psi - \operatorname{cosec}\psi\cdot\cot\psi]$$

$$= -a\cdot\cot^2\psi\cdot\operatorname{cosec}\psi - a\operatorname{cosec}^3\psi - \frac{a\operatorname{cosec}\psi[\operatorname{cosec}\psi+\cot\psi]}{(\cot\psi+\operatorname{cosec}\psi)}$$

$$= -a\cot^2\psi\cdot\operatorname{cosec}\psi - a\operatorname{cosec}^3\psi - a\operatorname{cosec}\psi$$

$$= -a(\operatorname{cosec}^2\psi - 1)\operatorname{cosec}\psi - a\operatorname{cosec}^3\psi - a\operatorname{cosec}\psi$$

$$= -a\operatorname{cosec}^3\psi + a\operatorname{cosec}\psi - a\operatorname{cosec}^3\psi - a\operatorname{cosec}\psi$$

$$= -2a\operatorname{cosec}^3\psi.$$

$\therefore |\rho| = 2a\operatorname{cosec}^3\psi$.

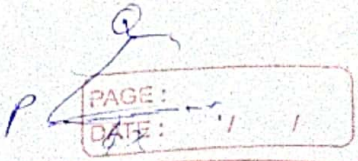
2. Find the radius of curvature for the semi-cubical parabola $s = \frac{4}{9}a(\sec^3\psi - 1)$ at the point (s, ψ).

→

$$\frac{ds}{d\psi} = \frac{4}{9}a[3\sec^2\psi\cdot\sec\psi\cdot\tan\psi]$$

$\therefore \rho = \frac{4}{3}a\sec^3\psi\cdot\tan\psi$.

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* Radius of curvature for cartesian Equations.

Consider the curve $y = f(x)$

We know that $s = \frac{ds}{d\psi}$.

Since, $\frac{dy}{dx} = \tan\psi$.

$$\therefore \frac{d^2y}{dx^2} = \sec^2\psi \frac{d\psi}{dx} = \sec^2\psi \frac{d\psi}{ds}\frac{ds}{dx}$$

$$\frac{d^2y}{dx^2} = \sec^2\psi \cdot \frac{d\psi}{ds} \quad \left(\because \frac{dx}{ds} = \cos\psi\right)$$

hypotenuse angle

will taking components along X-axis

Hence, the curvature $\frac{d\psi}{ds} = \frac{\frac{d^2y}{dx^2}}{\sec^2\psi}$

$$= \frac{\frac{d^2y}{dx^2}}{(1+\tan^2\psi)^{3/2}}$$

$$= \frac{\frac{d^2y}{dx^2}}{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}$$

- The radius of curvature at $P(x, y)$ is $s = \frac{ds}{d\psi}$.

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$$\frac{\left[1+\left(\frac{1}{\frac{dy}{dx}}\right)^2\right]^{3/2}}{\frac{1}{\left(\frac{dy}{dx}\right)}} = \frac{[1+x_1^2]^{3/2}}{x_2}$$

$$\therefore \rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

* Examples:-

1 Find the radius of curvature at any point of the catenary $y = c \cdot \cosh\frac{x}{c}$ and hence prove that $\rho = \frac{y^2}{c}$.

→

$$y = c \cdot \cosh\frac{x}{c}$$

$$\frac{dy}{dx} = \left(c \cdot \sinh\frac{x}{c}\right) \cdot \frac{1}{c}$$

$$= \sinh\frac{x}{c}$$

$$\frac{d^2y}{dx^2} = \frac{1}{c} \cdot \cosh\frac{x}{c}$$

$$\therefore \rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = \frac{[1+y_1^2]^{3/2}}{y_2}$$

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$$\rho = \frac{\left[1 + \sinh^2 \frac{x}{c}\right]^{3/2}}{\frac{1}{c}\cosh \frac{x}{c}} \qquad \left[\cosh^2 x - \sinh^2 x = 1\right]$$

$$= c \cdot \frac{\cosh^3 \frac{x}{c}}{\cosh \frac{x}{c}}$$

$$\rho = c \cosh^2 \frac{x}{c}$$

$$\frac{\rho}{c} = \cosh^2 \frac{x}{c}$$

$$= \left(\frac{y}{c}\right)^2$$

$$\rho = \frac{y^2}{c}.$$

2. Find the radius of curvature of $y = 4\sin x - \sin 2x$ at $x = \frac{\pi}{2}$.

→

$$y = 4\sin x - \sin 2x$$

$$\frac{dy}{dx} = 4\cos x - 2\cos 2x$$

$$\frac{d^2y}{dx^2} = -4\sin x + 4\sin 2x$$

$$\left(\frac{dy}{dx}\right)_{x=\frac{\pi}{2}} = 4\cos\frac{\pi}{2} - 2\cos 2\left(\frac{\pi}{2}\right)$$

$$= 0 - 2(-1)$$

$$= 2$$

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$$\left(\frac{d^2y}{dx^2}\right)_{x=\frac{\pi}{2}} = -4\sin\frac{\pi}{2} + 4\sin 2\left(\frac{\pi}{2}\right)$$

$$= -4 + 4\sin\pi$$

$$= -4 + 0$$

$$= -4$$

$$\therefore \rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

$$= \frac{[1+4]^{3/2}}{-4}$$

$$= \frac{5^{3/2}}{-4}$$

$$|\rho| = \frac{5\sqrt{5}}{4}$$

Ex. Find the R.O.C. at $xy^2 = (x+4)$ at $(-4, 0)$.

→ here, in $\rho = \left[\frac{(1+y_1^2)^{3/2}}{y_2}\right]$ but y_1 is infinite. so will take, $\rho = \left[\frac{(1+x_1^2)^{3/2}}{x_2}\right]$ Now,

$$\frac{dx}{dy} = x_1 \Rightarrow 2x + x_1y^2 = 16x_1$$

$$x_1 = \frac{2xy}{16-y^2} = \frac{2(-4)(0)}{16-(0)^2} = 0$$

$$\frac{d^2x}{dy^2} = x_2 \Rightarrow 2x + 2x_1y = (16-y^2)x_2 - 2yx_1$$

$$x_2 = \frac{2x}{16-y^2} = \frac{-8}{16} = -\frac{1}{2}.$$

$$\rho = \frac{(1+(0)^2)^{3/2}}{\left(-\frac{1}{2}\right)} = \frac{1^{3/2}}{-\frac{1}{2}} \Rightarrow \rho = -2 \Rightarrow \boxed{|\rho| = 2}$$

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20.9 Radius of curvature at the origin

In order to find the radius of curvature at the origin of the curve $y = f(x)$, we shall discuss following two methods.

(1) Newton's method (2) Series method.

(1) Newton's method

(i) If x–axis is tangent to the curve $y = f(x)$ at the origin, then

$$\rho_{at\ (0,\ 0)} = \lim_{\substack{x \to 0 \\ x \to 0}} \frac{x^2}{2y}$$

(ii) Similarly, if y–axis is tangent to the curve $y = f(x)$ at the origin, then

$$\rho_{at\ (0,\ 0)} = \lim_{\substack{x \to 0 \\ y \to 0}} \frac{y^2}{2x}$$

(2) Series method

If neither x–axis nor y–axis is tangent at the origin, we use Maclaurin's series.

$$y = f(x) = f(o) + xf'(0) + \frac{x^2}{2!} f''(0) + \ldots$$

But $f(0) = 0$, let $f'(0) = p$ and $f''(0) = q$ then

$$y = px + q\frac{x^2}{2!} + \ldots$$

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Putting this value of y in the equation of the curve $y = f(x)$ and equating to zero the coefficients of various powers of x, we get values of p and q.

Then, $\rho_{at\ (0,\ 0)} = \dfrac{(1 + p^2)^{\frac{3}{2}}}{q}$

EXAMPLE 1 Find the radius of curvature at the origin for the curve $y^4 + x^3 + 3x^2 + 3y^2 - 9y = 0$.

SOLUTION Equating to zero, the lowest degree term i.e. $-9y = 0$, we get $y = 0$.

$\therefore$ x–axis is the tangent at the origin.

Now, dividing by y we get,

$$y^3 + x\left(\frac{x^2}{y}\right) + 3\left(\frac{x^2}{y}\right) + 3y - 9 = 0$$

Now, taking the limit as $x \to 0$, $y \to 0$ and putting $\dfrac{x^2}{y} = 2\rho$,

we get $0 + 0.\ 2\rho + 3.2\ \rho + 0 - 9 = 0$

$\therefore \quad \rho = \dfrac{3}{2}$

EXAMPLE 2 Find the radius of curvature at the origin for the curve $y^2 - 3xy - 4x^2 + x^3 + x^4y + y^5 = 0$

SOLUTION Equating to zero, the lowest degree terms,

we get $y^2 - 3xy - 4x^2 = 0$

or $(y - 4x)(y + x) = 0$

$\Rightarrow \quad y = 4x$ and $y = -x$

$\therefore \quad y = 4x$ and $y = -x$ are two tangents at the origin.

Putting $y = px + \dfrac{1}{2}qx^2 + \ldots$ in the given equation, we get

$$\left(px + \frac{qx^2}{2} + \ldots\right)^2 - 3x\left(px + \frac{qx^2}{2} + \ldots\right) - 4x^2 + x^3$$

$$+ x^4\left(px + \frac{qx^2}{2} + \ldots\right)^5 = 0$$

Equating to zero the coefficients of x^2 and x^3 we get,

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$p^2 - 3p - 4 = 0$ and $pq - \frac{3}{2}q + 1 = 0$

$\therefore$ $(p - 4)(p + 1) = 0$ $\quad \therefore \quad p = 4, -1$

when $p = 4$, $\quad q = -\frac{2}{5}$

and when $p = -1$, $\quad q = \frac{2}{5}$

$$\therefore \quad \rho = \frac{\left(1 + p^2\right)^{\frac{3}{2}}}{q} = \frac{(1 + 16)^{\frac{3}{2}}}{-\frac{2}{5}} = -\frac{85}{2}\sqrt{17} \text{ and}$$

$$\rho = \frac{(1 + 1)^{\frac{3}{2}}}{\frac{2}{5}} = 5\sqrt{2}.$$

Hence the radius of curvature at the origin are $-\frac{85\sqrt{17}}{2}$ and $5\sqrt{2}$.

EXERCISES 20.8

Find the radius of curvature at the origin for

1. $y^4 + x^3 + a(x^3 + y^3) - a^2y = 0$
2. $y = x + x^2 + 2xy + y^2$

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